

**EMPLOYEE
ATTRITION
PREDICTION USING
RANDOM FOREST
ON**

Employee Attrition Prediction Using Random Forest Algorithm

project report submitted in partial fulfilment of the requirements for the Course

**MACHINE LEARNING
BACHELOR OF TECHNOLOGY**

in

A.Y.: 2026-27

by

Name Ch. Shravya chowdary roll no:2520030322

Under the Esteemed guidance of

Shaik Asif

Assistant professor, CSE

CSE



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

Off-Campus: Bachupally-Gandimaisamma Road, Bowrampet, Hyderabad, Telangana - 500 043.

Phone No: 7815926816, www.klh.edu.in

DEPARTMENT OF _____

Name of Title: Employee Attrition Prediction Using Random Forest Algorithm

1. Abstract:

Employee attrition is a major concern for organizations because the unexpected departure of employees can negatively affect productivity, team performance, and overall business growth. Replacing employees also involves significant recruitment, training, and onboarding costs. Therefore, predicting employee attrition in advance can help organizations take appropriate actions to improve employee retention. This project, Employee Attrition Prediction Using Random Forest Algorithm, focuses on developing a machine learning-based system to predict whether an employee is likely to leave an organization.

The proposed system analyses various employee-related factors, including age, salary, job satisfaction, overtime, years of experience, work-life balance, job role, department, and working environment. The collected employee dataset is first pre-processed by handling missing values, converting categorical information into numerical form, and selecting relevant features. The processed data is then divided into training and testing datasets. A Random Forest classification algorithm is applied to identify patterns and relationships between employee characteristics and attrition.

Random Forest is selected because it combines multiple decision trees to produce more reliable and accurate predictions while reducing the risk of overfitting. The performance of the model can be evaluated using metrics such as accuracy, precision, recall, F1-score, and confusion matrix. The trained model can classify employees into two categories: those who are likely to stay and those who are at risk of leaving.

The proposed system can provide organizations with useful insights into the major factors influencing employee attrition. By identifying employees who may be at higher risk of leaving, management can take preventive measures such as improving job satisfaction, managing workloads, providing career development opportunities, and offering suitable incentives. Overall, this project demonstrates how machine learning and predictive analytics can support data-driven decision-making and help organizations develop effective employee retention strategies.

2. INTRODUCTION:

Employee attrition refers to the process of employees leaving an organization voluntarily or involuntarily. It is a common challenge faced by companies across different industries, as high employee turnover can affect productivity, team performance, and organizational growth. Recruiting and training new employees also requires considerable time and financial resources. Therefore, predicting employee attrition in advance can help organizations take effective measures to retain valuable employees.

With the growth of Machine Learning, organizations can analyze employee data and identify patterns that may indicate a possibility of an employee leaving. Factors such as salary, job satisfaction, age, overtime, experience, job role, work-life balance, and working environment can influence an employee's decision to stay or leave.

This project, Employee Attrition Prediction Using Random Forest Algorithm, aims to develop a machine learning model that predicts the likelihood of employee attrition. The Random Forest algorithm is used because it can handle multiple features effectively and provide reliable classification results.

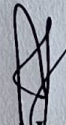
The proposed system analyses historical employee data, preprocesses the information, trains the Random Forest model, and predicts whether an employee is likely to leave the organization. The results can help management identify potential attrition risks and take suitable actions to improve employee satisfaction, reduce turnover, and maintain a stable workforce.

3. SOFTWARE REQUIREMENTS:

- Operating System: Windows / Linux / macOS
- Programming Language: Python 3.x
- IDE: Jupyter Notebook / Google Colab / VS Code
- Libraries:
 - Pandas – Data processing
 - NumPy – Numerical operations
 - Scikit-learn – Random Forest model and evaluation
 - Matplotlib – Data visualization
 - Seaborn – Graphs and visualization
- Dataset: Employee Attrition Dataset (CSV)
- Web Browser: Chrome / Edge / Firefox



Signature of the Guide



Course Instructor